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Research note

A palaeomagnetic survey of the Jotnian dolerites of central–east Sweden

J. D. A. Piper *Sub-Department of Geophysics, University of Liverpool, PO Box 147, Liverpool L69 3BX*

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Summary. The Jotnian igneous episode of central–east Sweden is represented by dolerite sills distributed over an area of 5000 km^2. After detailed af and thermal cleaning of 265 samples from 46 sites in these dolerites, 43 of the sites yield a mean direction of magnetization $D = 47.8°$, $I = -45.5°$ ($\alpha_{95} = 2.6°$) and a palaeomagnetic pole at 156.5° E, 7.5° S. The results are combined with other data from the same region to derive an overall mean palaeomagnetic pole for the Jotnian igneous episode. All samples have the same polarity and directions of magnetization agree closely with dolerites attributed to the Jotnian episode from elsewhere in the Baltic Shield. The Jotnian igneous intrusions of Sweden are dated 1247–1213 Ma and are broadly contemporaneous with the Mackenzie magmatic episode in North America (1205–1189 Ma) and dyke intrusions within the Gardar Province of South Greenland (1250–1212 Ma). These igneous events all have the same polarity on the continental reconstruction derived from 2000–1200 Ma palaeomagnetic data with the Laurentian and Baltic Shields in juxtaposition.

1 Introduction

Throughout the Baltic Shield unaltered and undeformed dolerites are widely distributed in an intrusive relationship to Proterozoic sediments collectively grouped together as the 'Jotnian Formation' (Sederholm 1927). The dolerites (although widely referred to in the literature as 'Jotnian', they should strictly be defined as 'post-Jotnian') have been dated 1245 Ma in central Sweden by the K–Ar isochron method (Welin & Lundqvist 1975), a determination recently confirmed by extensive Rb–Sr studies of Patchett, Byland & Upton (1978). The general objectives of palaeomagnetic work on these dolerites are firstly to determine whether they all represent a single major igneous event covering one or more polarity episodes, and secondly to establish a palaeomagnetic pole or poles linked to the radiometric studies. The present paper reports the results of a detailed sample of the dolerites covering the outcrops in central–east Sweden (Fig. 1).

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2 Geology

The Jotnian doleries intrude Svecofennian basement rocks (> 1600 Ma) and the Rapakivi-type intrusion at Nordingrå (dated 1415 ± 30 Ma, Lundqvist & Samuelsson (1973)) as sheet-like bodies. At Nordingrå (18.3° E, 62.9° N) the dolerites are approximately conformable with the bedding of Jotnian sediments, and the major part of the whole dolerite outcrop has an oval shape (Fig. 1, Lundqvist & Samuelsson 1973) with a probable lopolithic form. The individual sills have shallow dips defining a centre close to the site of the older Nordingrå intrusion, but they mainly invade fractures in the basement and there is no

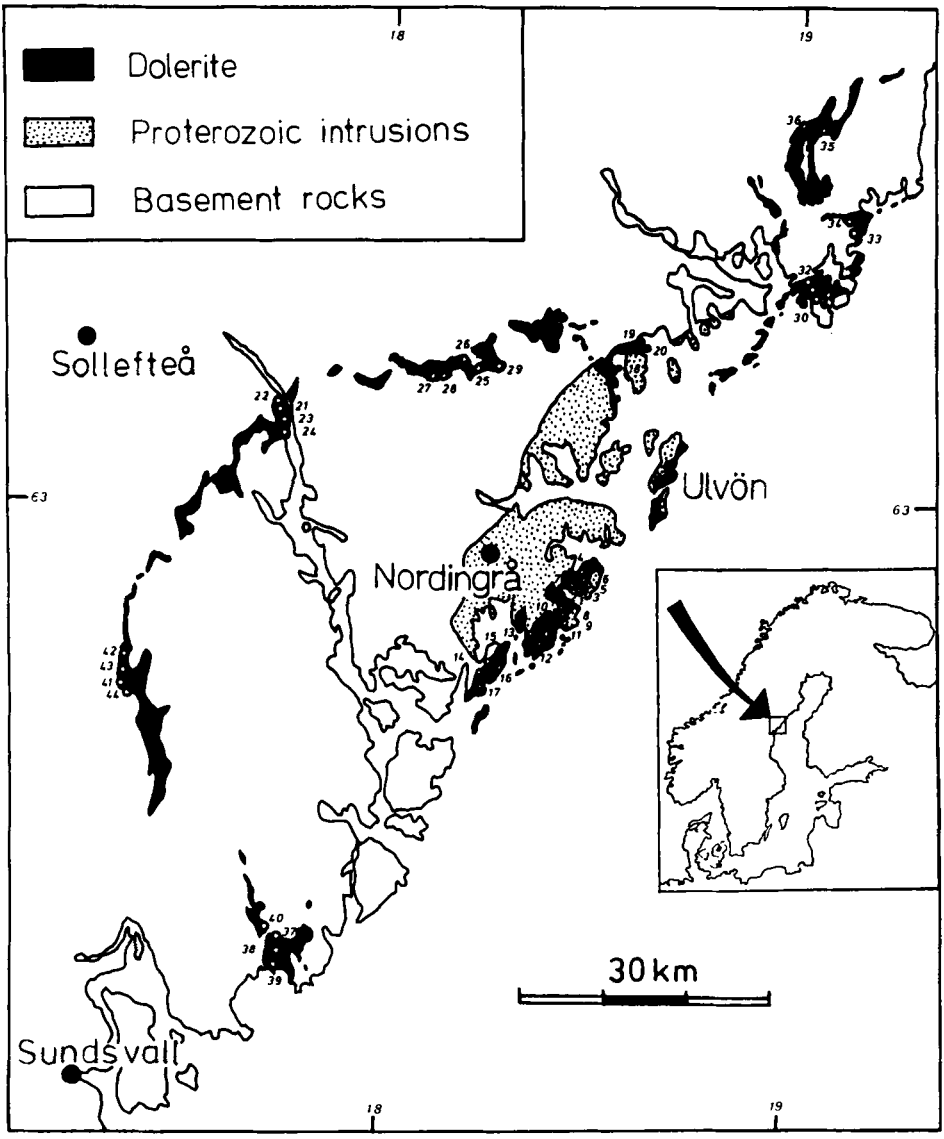


Figure 1. Outcrop of the Jotnian dolerites in central–east Sweden with palaeomagnetic sampling sites of this study; the outcrop map is after Welin & Lundqvist (1975). The two sites (45 and 46) not shown on this diagram come from a road cut through the sill north of Gnarp at 17.3° E, 62.1° N, and are separated by approximately 0.5 km. The inset diagram shows the regional location of this area within Scandinavia.

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evident regional tilt relevant to the palaeomagnetic study. They are rhythmically layered and mostly range from 200–500 m in thickness, but may reach 800 m in thickness at some localities (Magnusson & Larson 1977). The intrusions have an alkali–olivine basalt composition with plagioclase, pyroxene, olivine and biotite, and 5 per cent or less Ti–Fe oxides (Welin & Lundqvist 1975). Larson & Magnusson (1976) report magnetite, ilmenite and ulvöspinel as the major opaque phases with the ilmenite occurring predominantly as exsolution lamellae in host magnetite; these authors also note magnetite in host olivine crystals to the extent of up to 5 per cent of the crystal volume, and also possibly present as rods in pyroxenes.

3 Palaeomagnetic studies

The dolerite intrusions embraced by this study are mostly rather poorly exposed with outcrops confined to road cuts, the Baltic shore and certain glaciated hill slopes. Samples were collected in the form of cores using a portable drill and oriented by sights on the Sun, or, in a few cases, using a magnetic compass. In the laboratory all samples were progressively demagnetized in alternating fields in steps of 5 mT (50 Oe) up to 30 mT and then in steps of 10 mT to peak fields ranging from 60 to 150 mT.

Behaviour to this treatment was highly systematic: all samples started with an initial shallow negative inclination and moved towards a steeper negative inclination with progressive treatment (Fig. 2). This movement clearly reflects the removal of a small viscous component of remanence which is probably associated with the present field because NRM intensities often exhibit a slight increase over the first few steps, and demagnetization usually follows a path close to a great circle passing through the present geomagnetic field

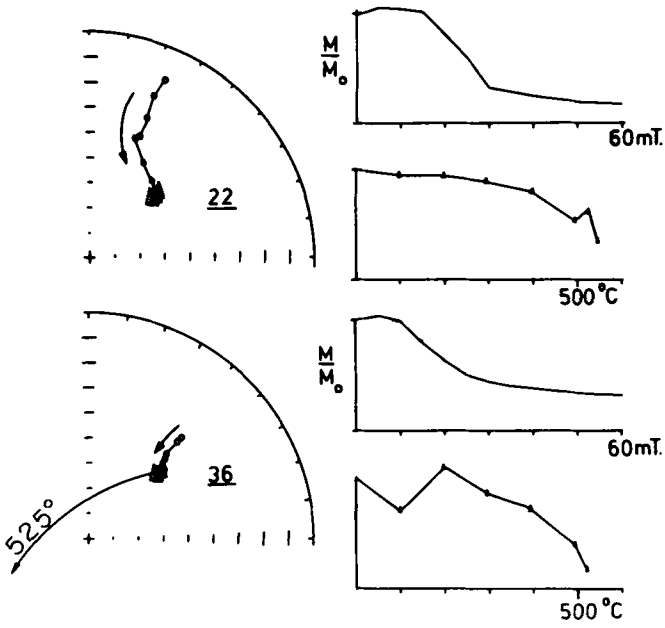


Figure 2. Two examples of progressive af (circles) and thermal (triangles) demagnetization of NRMs in Jotnian dolerites. Intensity of magnetization is expressed as a fraction of the initial moment (M_0); closed symbols are lower hemisphere plots and open symbols are upper hemisphere plots.

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Table 1. Jotnian dolerites of central–east Sweden, site mean palaeomagnetic statistics after af and thermal cleaning.

Site no.	N	D	I	R	k	α_{95}	M_0 ($\times 10^{-5}$ A.m^2/kg)
1	7	49.8	−45.3	6.97	234	3.9	4.59–2.61
2	7	43.3	−49.0	6.98	343	3.3	7.49–3.57
3	6	35.3	−53.1	5.68	16	17.5	2.55–0.13
4	6	50.6	−48.2	5.98	318	3.8	16.98–3.62
5	5	49.1	−45.5	4.99	626	3.1	20.23–7.33
6	6	54.6	−54.9	5.98	259	4.2	13.94–9.96
7	5	42.7	−37.0	4.95	82	8.5	2.72–1.92
8	5	48.7	−37.1	4.88	33	13.5	6.61–4.64
9	6	44.0	−42.7	5.97	174	5.1	14.43–10.63
10	5	50.5	−47.9	4.95	85	8.3	6.41–2.44
11	6	48.7	−37.9	5.90	50	9.6	15.23–7.32
12	6	48.5	−36.6	5.90	48	9.8	11.93–5.02
13	6	48.1	−55.7	5.97	162	5.3	5.17–0.80
14	6	42.7	−36.4	5.99	419	3.3	4.61–3.58
15	6	49.6	−31.4	5.97	171	5.1	3.71–2.65
16	5	44.5	−34.7	4.99	772	2.8	13.32–9.35
17	6	46.7	−45.5	5.97	157	5.4	10.52–4.10
18	6	54.0	−28.5	5.94	80	7.5	33.28–6.51
19	6	48.0	−47.9	5.99	685	2.6	3.00–1.56
20	6	51.6	−63.0	5.97	172	5.1	4.12–2.42
21	7	43.9	−51.6	6.99	960	1.9	2.84–1.93
22	6	47.7	−56.1	5.90	53	9.3	1.87–0.80
23	6	35.0	−33.1	5.99	473	3.1	2.84–2.43
24	6	52.4	−43.0	5.93	76	7.8	4.33–2.86
25	5	45.7	−27.9	4.55	9	27.0	22.86–0.11
26	5	46.5	−51.1	4.98	218	5.2	2.78–2.02
*27	4	70.0	10.8	3.99	279	5.5	8.56–3.87
*28	6	48.6	−75.9	5.94	77	7.7	10.42–4.01
*29	3	101.7	3.0	2.99	7014	1.5	8.01–3.23
30	6	58.6	−40.8	5.98	247	4.3	7.68–4.86
31	6	55.4	−41.0	5.99	860	2.3	5.80–3.94
32	6	49.0	−37.4	5.99	2560	1.3	3.93–1.77
33	6	35.4	−37.0	5.96	136	5.8	2.68–1.54
34	6	44.6	−30.9	5.98	315	3.8	1.98–1.06
35	6	46.7	−54.0	5.96	140	5.7	1.69–0.87
36	5	42.9	−52.7	4.99	1321	2.1	2.77–1.30
37	6	48.7	−44.0	5.99	739	2.5	2.32–1.33
38	6	47.0	−53.0	5.99	572	2.8	2.22–1.06
39	6	51.1	−54.8	5.96	118	6.2	1.49–1.01
40	5	54.3	−56.6	4.99	589	3.2	4.03–1.32
41	6	45.6	−57.1	5.98	234	4.4	1.28–0.81
42	6	46.1	−46.3	5.97	193	4.8	6.89–2.31
43	6	46.1	−51.8	5.99	334	3.7	0.68–0.46
44	6	46.6	−53.7	5.95	98	6.8	1.98–1.58
45	6	59.4	−49.1	5.94	86	7.3	3.31–1.55
46	6	60.2	−46.2	5.98	273	4.1	4.03–1.89
Combined mean direction (43 sites):							Palaeomagnetic pole
		47.8	−45.5	42.4	71	2.6	156.5° E, 7.5° S (dp = 2.1°, dm = 3.3°)

* Sites excluded from group mean calculation. D = declination and I = inclination of N sites or samples with resultant vector R, precision $k(=(N-1)/(N-R))$ and 95 per cent confidence circle radius α_{95}; M_0 is the total NRM intensity, dp and dm are the radii of the 95 per cent confidence oval about the mean pole in the colatitude direction and at right angles respectively.

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direction (Fig. 2). The limit of this systematic movement was achieved at the 20–30 mT stage of treatment for most samples, but not until the 60–70 mT stage for a few samples; at higher alternating fields, sample directions either remained at these end points or showed increasingly erratic movements associated with the introduction of unstable components by the af procedure. The progressive af treatment of all samples, although more time-consuming than the selection of a single representative cleaning field, was justified by the range of fields at which this viscous component was fully removed, and by the very high within-site precisions (k), sometimes exceeding values of 1000, which have been achieved at most localities (Table 1).

The af characteristics of this collection are essentially very similar to those reported from other studies of the same rock suite (Poorter 1976; Larson & Magnusson 1976). Also, in studies of Jotnian dolerites from Finland (Neuvonen 1965, 1966) the overall mean direction computed from the af cleaned directions has a steeper negative magnetic inclination than that computed from the total NRM, and reflects the removal of a similar small viscous component to that present in the Swedish dolerites.

To determine whether the stable end points defined by the af cleaning relate to the initial igneous episode and reside in single high blocking temperature components, one pilot sample from each site has also been thermally demagnetized at a range of steps up to 525° C. The residual magnetic moments in all samples do not show appreciable fall until temperatures approaching the Curie points, and directions of magnetization remain identical to the af cleaned directions (Fig. 2). It is inferred that the af cleaning treatment has successfully recovered the primary magnetization in these dolerites; the simple behaviour to stages of this treatment is consistent with the opaque mineralogy, as neither maghemite nor hematite have been identified in these rocks (Larson & Magnusson 1976).

In an early study of Jotnian dolerites Mulder (1971) considered that they may have been remagnetized in Palaeozoic times but this possibility has been rejected by later authors, notably by Poorter (1976). In addition to the evidence documented by this latter author, the Jotnian dolerites in east Sweden are remote from the Caledonian Front, they have partially remagnetized the older Nordingrå Complex and also have a magnetization quite different from the (Cambrian) Alnö Island Alkaline Complex in the same area (Piper, in preparation).

From measurements on magnetite concentrates from the Ulvö region (18.7° E, 68.0° N) Larson & Magnusson derived Curie points close to that of pure magnetite. Thermomagnetic determinations on whole rock samples (e.g. Fig. 3) also yield single Curie points in the range 500–575° C with little or no change on heating in air, and suggest that the host magnetite has achieved a composition near to that of pure magnetite by exsolution of ilmenite lamellae (Ade-Hall, Palmer & Hubbard 1971); no hematite has been identified from these investigations.

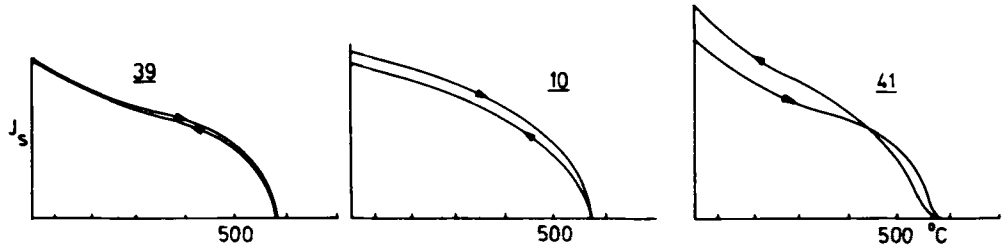


Figure 3. Three typical thermomagnetic determinations on whole rock samples of Jotnian dolerites; the saturation magnetization (J_s) axis is arbitrary.

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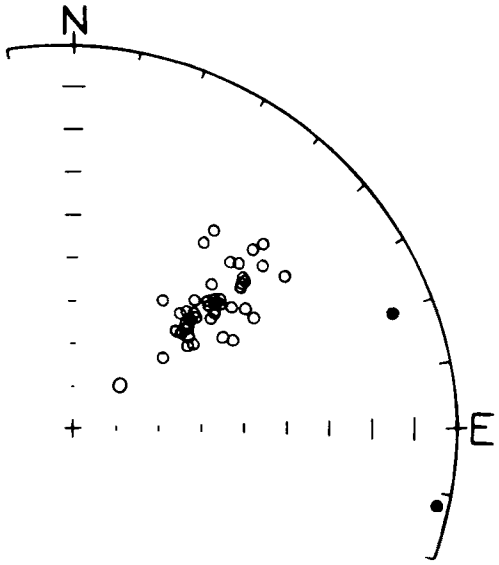


Figure 4. A stereographic plot of all site mean directions of magnetisation from this study after af and thermal cleaning.

All site mean directions of magnetization with three exceptions (27–29) lie within a narrow and similar range (*ca.* 30°) of both declination and inclination (Fig. 4) which is further evidence for the success of the cleaning techniques. The mean of these 43 sites is $D = 47.8°$, $I = -45.5°$ ($\alpha_{95} = 2.6°$). Although mineral layering with dips up to 25° is present at some sites (and notably well exposed along the Baltic shore) this is clearly a primary igneous feature and tilt adjustment destroys the close agreement of site mean directions illustrated in Fig. 4; also, as anticipated, there is no significant difference in site mean directions from the east and west limbs of the lopolithic body. The three anomalous sites (27, 28 and 29) all come from a traverse across the sill at 18.1° E, 63.2° N. Magnetic remanence is relatively high at these localities (Table 1) and it is not obvious that they record transitional field directions which are usually associated with low values of geomagnetic field intensity; geomagnetic field intensities may, however, reach high values for short periods of the field transition (Shaw 1975).

The palaeomagnetic pole derived from the mean direction is at 156.6° E, 7.5° S, and the mean pole position of all virtual geomagnetic poles (the sampled localities range from 62.0 to 63.4° N and 17.3 to 19.1° E) is at 156.6° E, 8.1° S.

4 Discussion

4.1 MEAN PALAEOMAGNETIC POLE FOR THE JOTNIAN DOLERITES OF CENTRAL–EAST SWEDEN

The present study supplements and greatly extends some earlier palaeomagnetic work on the Jotnian dolerites in eastern Sweden. Poorter (1976) reported af cleaned results from three sites; two of these are near sites 11–13 of this study and the other is close to sites 45 and 46. The directions of magnetization reported by Poorter are very similar to those isolated by the present study after similar treatment (Table 2). Magnusson & Larson (1977) report af cleaned results from 32 block samples of which many are derived from the Baltic Islands forming the eastern margin of the outcrop illustrated in Fig. 1; they also (Larson &

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Table 2. Jotnian dolerites of east–central Sweden – summary of all palaeomagnetic results and overall mean.

Reference	Sampled localities	Sites/ Samples	$D°$	$I°$
Magnusson & Larson (1977)	Samples derived mainly from coastal exposures between 62.9 and 63.3° N	32/32	40.0 (after cleaning at 20 mT)	−35.8*
Larson & Magnusson (1976)	Samples derived from island of North Ulvön and mainland near sites 1–4 of this study	20/20	50.4 (after cleaning at 30 mT)	−38.7*
Poorter (1976)	Samples collected 8 km (1) south-east of Nordingrå (2) 10 km north of Gnarp (3)	1/6	45.8	−52.5
		1/6	45.4	−32.4
		1/6	45.8	−51.8
This study	Samples distributed over whole dolerite outcrop	43/252	47.8	−45.5
Mean of all studies:		48/322	47.5	−45.1
		Palaeomagnetic pole:		
(R = 47.3, k = 71, α_{95} = 2.5°)		156.9° E, 7.2° S (dp = 2.0°, dm = 3.1°)		

* Selected for group means showing highest overall precisions.

Magnusson 1976) give results for 20 block samples derived mainly from the island of north Ulvön. The mean directions for these studies are listed in Table 2. The present study increases the number of palaeomagnetic samples from these dolerites by more than four times and vastly increases the areal coverage. It is now possible to conclude with some certainty that the dolerites are all of one polarity, and the collective earlier results are combined in Table 2 with the 43 sites of the present study to derive an overall mean for the Jotnian dolerites of this general region (D = 47.5°, I = −45.1°, α_{95} = 2.5°). Unfortunately this computation does not do full justice to the wide geographic distribution of the samples of Magnusson & Larson (1977) for which these authors quote a combined mean direction; since, however, their results are based on single block samples from each locality they are not truly comparable with the site means given in this study, Larson & Magnusson (1976), and Poorter (1976).

4.2 AGE AND AREAL DISTRIBUTION OF THE JOTNIAN IGNEOUS EVENT

Welin & Lundqvist (1975) derived a K–Ar isochron age of 1245 ± 20 Ma from samples taken from three parts of the outcrop illustrated in Fig. 1. This age has been adjusted to 1254 Ma by Patchett *et al.* (1978) using the new internationally agreed decay constants (Steiger & Jäger 1977). Patchett *et al.* summarize Rb–Sr dates related to the widely distributed *c.* 1200 Ma dolerites of the Baltic Shield, and 13 of the 19 ages (Rb–Sr, U–Pb or K–Ar isochron) relating to these events lie in the range of 1270–1213 Ma. The dolerites for which both palaeomagnetic and radiometric data are now available are summarized in Table 3; the new data of Bylund listed in Patchett *et al.* (1978) comes from an area originally studied by Mulder (1971); the four sites from this latter study are excluded from Table 2 because of the poor agreement between them (magnetic declination ranges from 340° to 31° E and magnetic inclination from +15° to −42°). Poles 1 to 4 are first rank

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Table 3. A compilation of palaeomagnetic and radiometric evidence relating to the *c.*1200 Ma dolerites of Scandinavia.

Rock unit(s)	Radiometric age/method		Age ref.	Pole position	*dp/dm*	Pde ref.
[1] Finland						
(1) Vaasa	1225 ± 18, 1270,	Rb–Sr U–Pb	Patchett *et al.* (1978)	164° E, 7° N	6.0/3.0	Neuvonen (1976)
(2) Satakunta	1236 ± 18, 1225 ± 18,	Rb–Sr Rb–Sr	Patchett *et al.* (1978)	158° E, 2° N	4.8/2.8	Neuvonen (1965)
(3) Märket	1270,	U–Pb	Kouvo (1976)	146° E, 6° S	11.0/7.0	Neuvonen & Gundström (1969)
[2] Sweden						
(4) Vasternorrland	1254 ± 20	K–Ar (isochron)	Welin & Lundqvist (1975)	157° E, 7° S	2.0/3.1	This paper
(5) Gävle	–	–	–	150° E, 7° N	(VGP)	Poorter (1976)
(6) Älvdalssen	1231 ± 18,	Rb–Sr	Patchett *et al.* (1978)	149° E, 1° S	(VGP)	Bylund, listed in Patchett *et al.* (1978)
(7) Alvho	1290 ± 63 1215 ± 18	Rb–Sr Rb–Sr	Patchett *et al.* (1978) Patchett *et al.* (1978)	170° E, 17° N	(VGP)	Bylund, listed in Patchett *et al.* (1978)
(8) Sundsjö and Gimån	1213 ± 18, 1156 ± 15 1167 ± 19 1179 ± 46 1229 ± 19	Rb–Sr Rb–Sr Rb–Sr Rb–Sr Rb–Sr	Patchett *et al.* (1978)	147° E, 3° N	(VGP)	Bylund, listed in Patchett *et al.* (1978)
(9) Dalarna	1200	K–Ar	–	115° E, 10° S	See text	Dyrelius (1978)

VGP = virtual geomagnetic pole.

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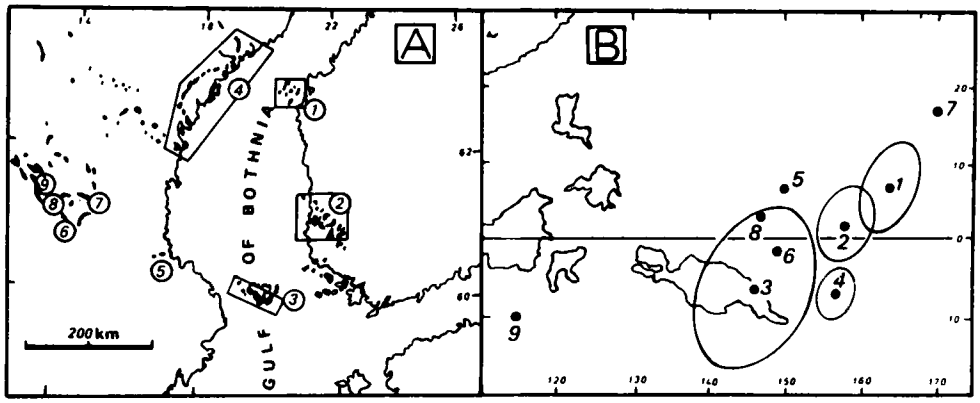


Figure 5. (a) The generalized distribution of Jotnian dolerites (black) in the Baltic Shield with the locations of palaeomagnetic studies numbered as they are listed in Table 3. (b) The palaeomagnetic poles with 95 per cent confidence ovals and virtual geomagnetic poles derived from the studies indicated in (a).

data based on large collections, while poles 5 to 9 are ancillary data based on isolated sites. Areal distribution of the data is shown in Fig. 5(a) and the distribution of palaeomagnetic poles in Fig. 5(b). With the exception of pole 7 which is regarded as a virtual geomagnetic pole only, and pole 9, the results from the separate Jotnian igneous events of Finland and Sweden are in close agreement with one another. In addition, all of these separate studies yield similar directions of magnetization of the same polarity. The reason why pole 9 is oblique to the remainder is uncertain; it is based on nine block samples from different sites but since cleaning was apparently only undertaken to 20 mT and the mean is very poorly defined, it does not fall in the same class as results 1–4. The results from Jotnian dolerites in Finland are given by Neuvonen (1965, 1966) and Neuvonen & Gundström (1969) and come from the Satakunta dykes and sills (18 sites, $D = 46°$, $I = -34°$), the Vaasa dykes (15 sites, $D = 38°$, $I = -29°$) and the Märket dykes (nine sites, $D = 53°$, $I = +12°$). It is inferred that either the Jotnian igneous episode lasted a shorter period of time than is suggested by the range of age dates, or that the position of the Baltic Shield and the geomagnetic field polarity were both constant for a long interval at *c.* 1270–1200 Ma.

4.3 SIGNIFICANCE OF THE *c.* 1200 MA IGNEOUS EPISODE

Many authors have noted that the Jotnian igneous events are broadly contemporaneous with the widespread *c.* 1200 Ma intrusion of dolerite dyke swarms in the Laurentian Shield. In the North American sector of this Shield, dyke swarms assigned to the Mackenzie igneous episode include the Mackenzie dykes and sills and the Sudbury dykes. Patchett *et al.* (1978) derive Rb–Sr ages in the range 1251–1189 Ma for these swarms which appear to be all of one polarity. In their palaeomagnetic study of this igneous province Fahrig, Gaucher & Larochelle (1965) reported results from 28 Mackenzie dykes from the southern part of the Canadian Shield and an unspecified number of Sudbury dykes. Fahrig & Jones (1969) later showed that 15 further sites in Mackenzie igneous rocks distributed between the northern part of the Shield and Hudson's Bay had the same polarity and yielded identical palaeomagnetic poles. The Gardar Igneous Province of South Greenland comprises a wide range of rocks formed between 1300 and 1500 Ma. Piper & Stearn (1977) and Piper (1977) find that east–west to east south-east–west north-west trending dolerite dykes of the

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Province yielding Rb–Sr ages of 1251–1212 Ma (Patchett *et al.* 1978) and the Kûngnât central complex (1219 ± 16 Ma) yield similar directions of magnetization of one polarity and similar palaeomagnetic poles to the Mackenzie igneous rock; these samples come from 37 sites distributed over 200 km^2. Basic igneous activity thus took place across most of the width of the Laurentian craton at *c.* 1250–1200 Ma during an interval of constant geomagnetic field polarity.

By comparing palaeomagnetic poles from the Baltic and Laurentian Shields, several workers have also shown that the two Shields were approximately in juxtaposition prior to 800 Ma although a quite different configuration is indicated by comparison of 2000–1200 Ma data (Piper 1976; Poorter 1976) and 1000–800 Ma data (Patchett & Bylund 1977; Morris & Roy 1977). The *c.* 1200–1000 Ma motion between the two shields is equivalent to a relative rotation of the Baltic Shield with respect to the Laurentian Shield, and has been genetically linked to the widespread Jotnian–Mackenzie–Gardar magnetism because this is the last known event to correlate with the 2000–1200 Ma reconstruction (Patchett *et al.* 1978). On this reconstruction, with the two shields in juxtaposition and the Grenville–Sveconorwegian mobile belts in alignment, the Jotnian igneous events have the same uniform polarity as the Mackenzie and Gardar events supporting the geochronologic evidence that these events were broadly synchronous.

Acknowledgments

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